

SANDUSTRY

Recipe Reference

Names as shown in game. A static snapshot - mods may add or replace recipes at runtime.

A chance below 100% is a yield, not a batch success rate: the input is consumed either way. Check the value table.

Contact Reactions

Two elements touching in the world convert on the spot. No machine required.

Water + Sand → Wet Sand

Water + Seed → Wet Seed

Water + Lava → Steam *lava persists*

Water + Flame → Steam

Void Petal + Redsand → Voidbloom

Shaker

Wet sand rides on top. Residue is left in place; gold falls through and out the underside.

Wet Sand → Residue + Gold 25% *gold drops 2 cells below*

Kinetic Press

Internally the velocity soaker. Input must be falling fast enough on impact.

Burnt Residue → Seed + Gold *needs downward velocity*

Planter Box

Wet seed germinates, the seedling runs, and where it blooms the terrain turns.

Wet Seed → Seedling *germinates in the box*

Seedling → Petal Soil + Gold Soil *on bloom, as terrain*

Smelter

Feed from above. Needs heat from an adjacent Thermal Buffer or lava.

Gold → Liquid Gold 50% *value-neutral: 0.5 x 2 = 1*

Copper → Liquid Copper

Condenser

Condenses gases to liquid. Needs cold from a Thermal Buffer or snow above.

Florin → Gold 50% or Florinol 50% *one or the other*

Steam → Water

Steam Dryer

Put Amethelis on top; steam underneath is the fuel.

Amethelis → Dry Amethelis

Snowmaker

Consumes water above and energy from the grid; snow spawns below.

Water → Snow costs energy, output below

Synthesizer

Built as the aurixiteCrystallizer. Fills a 4x4 pocket: sixteen units in, one crystal out.

Florinol → Aurixite 16 units + 800 energy

Burning

Set alight by flame, fire, a flamethrower, or a Burner Belt fuelled by a Thermal Buffer.

Residue → Burnt Residue 25% 75% is lost - intended

Dry Amethelis → Florin 50% florin rises as gas

Digging Terrain

What the ground gives up when excavated.

Dirt → Sand 50%

Grass → Sand 50%

Gold Soil → Gold

Petal Soil → Amethelis

Sporemound → Seed 50%

Frostbed → Snow 50% dig out before melting

Redsoil → Redsand 50%

Scoria → Cinder

Heat and Energy

What each machine costs to run, and why the Thermal Buffer beats raw lava.

Smelter	heat	Thermal Buffer on any side (-10 temp per smelt), or lava in the row directly below
Condenser	cold	Thermal Buffer (-2 temp per run), or snow above
Steam Dryer	steam	steam in the cells below, consumed as fuel
Snowmaker	energy	1 energy per tick, 100 ms interval
Synthesizer	energy	800 energy per conversion, all-or-nothing; queues and retries if short
Burner Belt	heat	Thermal Buffer adjacent; ignites flammables above. Without heat, a plain belt
Pyro Dispenser	heat	one flame burst per interval, drawn from an adjacent Thermal Buffer

Lava efficiency: the buffer wins 2.5 to 1

A Thermal Buffer absorbs one adjacent lava cell for +250 temperature, capped at 1000, and the lava is consumed. Each smelt drains 10, so one lava cell buys 25 smelts. Buffers also diffuse heat between themselves every second at rate 0.5, so one lava pit can feed a whole network. Lava placed directly under a smelter is consumed on 10% of smelts instead, which is only 10 smelts per cell. It only absorbs when there is headroom, so nothing is wasted.

Collectable Values

What a Collector pays. These are the only two valued elements in the build.

- Gold 1
- Liquid Gold 2

Smelting gold is value-neutral in expectation. The gain is density - one cell carries 2 instead of 1, halving entity count for the same money, and liquid gold moves through pipes. The cost is heat and variance.

Display Name to Internal ID

What you need when calling elements.getElementTypeFromId or structures.recipes.register.

Elements

Amethelis	petalium	Liquid Gold	liquidGold
Aurixite	aurixite	Redsand	sandium
Burnt Residue	burntResidue	Residue	residue
Cinder	basalt	Sand	sand
Copper	copper	Seed	seed
Dry Amethelis	dryPetalium	Seedling	seedling
Fire	fire	Snow	freezingIce
Flame	flame	Steam	steam
Florin	florin	Void Petal	voidPetal
Florinol	florinol	Voidbloom	gloom
Gold	gold	Water	water
Lava	lava	Wet Sand	wetSand
Liquid Copper	liquidCopper	Wet Seed	wetSeed

Terrains

Dirt	dirt	Petal Soil	Petal *
Frostbed	frostbed	Redsoil	redsoil
Gold Soil	GoldSoil *	Scoria	scoria
Grass	grass	Sporemound	sporemound

* No nameKey in the build - these terrains are unnamed in game; the label above is the internal constant.

Structures: Shaker=shaker Kinetic Press=velocitySoaker Planter Box=grower Condenser=thermofroster

Steam Dryer=thermodryer Synthesizer=aurixiteCrystallizer Thermal Buffer=thermalRelay Florinol Battery=goldBattery